$$z = \frac{x - \mu}{\sigma}$$

$$SR = \frac{\mu_R - \mu_{Rf}}{\sigma_R} \Rightarrow \frac{\text{risk premium}}{\sigma_R} \Rightarrow$$ want a big gap

$$R_p = w R + (1-w) R_f$$

mean ① $\mu_p = w \mu_R + (1-w) \mu_{Rf}$ " μ_f

$$= \mu_f + w \underbrace{(\mu_R - \mu_f)}_{\text{risk-premium}}$$

variance ② $\sigma_p^2 = w^2 \sigma_R^2 + (1-w)^2 \sigma_{Rf}^2 + 0$

$\sigma_{Rf}^2 \to = 0$ risk-free

$$\sigma_p = w \sigma_R$$

$$w = \frac{\sigma_p}{\sigma_R}$$

① $w = 0$ $R_p = R_f$

$\mu_p = \mu_{Rf}$

$\sigma_p = 0$

② $w = 1$ all my money on R

$\mu_p = \mu_R = 0.175$

$$\underbrace{\sigma_p = \sigma_R = 0.258}_{\text{Asset A slide 11}}$$

CAL slide 30

$$\mu_p = \underbrace{\mu_f}_{\text{intercept}} + \underbrace{\frac{\mu_R - \mu_f}{\sigma_R}}_{\text{SR slope.}} \sigma_p$$

Y

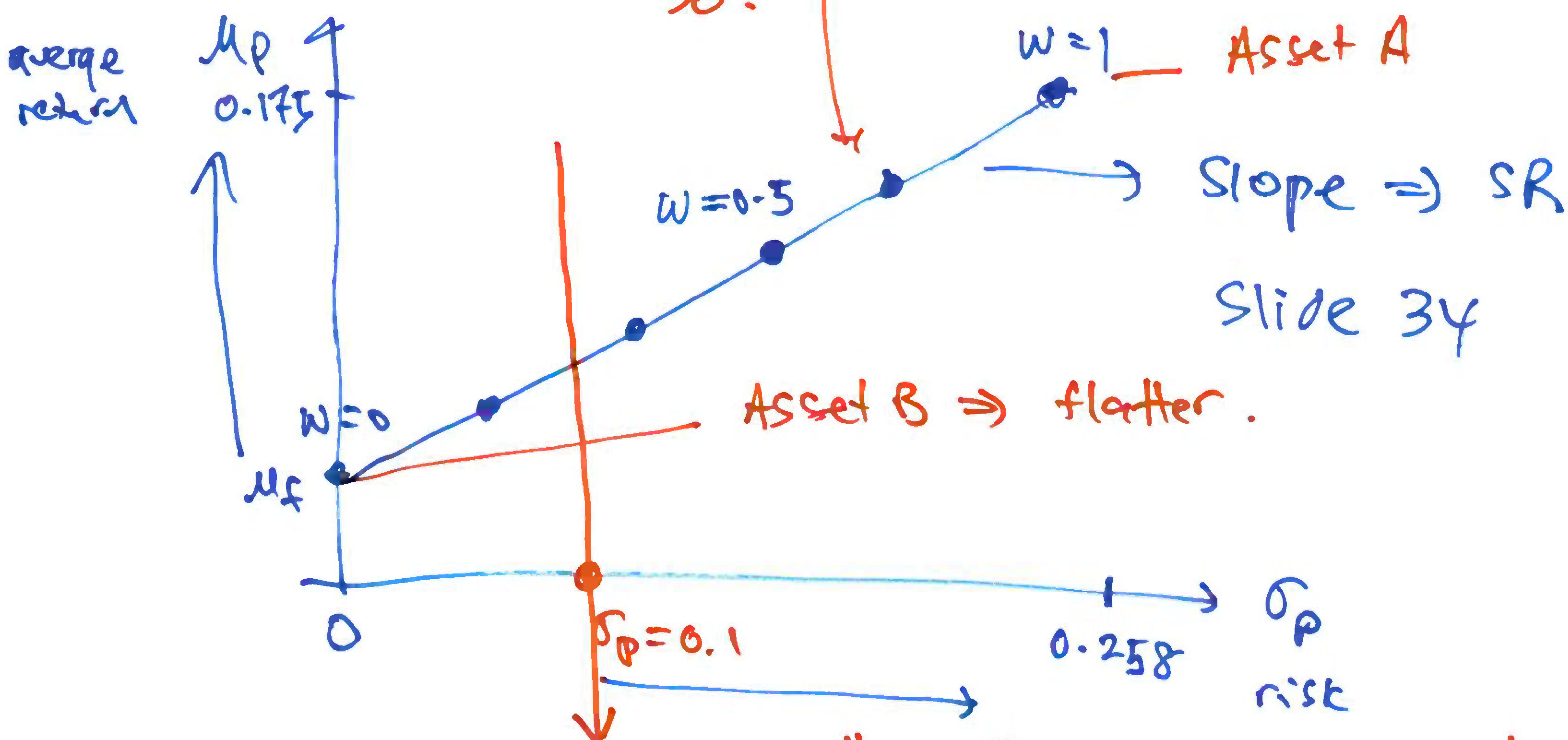
X

big gap btw μ_R and μ_f

SR bigger number

SR steeper slope.

slide 30.



higher return "port" A than B holding constant σ_p

$$\mathrm{var}(X+Y) = \mathrm{var}(X) + \mathrm{var}(Y) + 2\,\mathrm{cov}(X,Y)$$

$$\mathrm{cov}(X-Y, Z) = \mathrm{cov}(X,Z) - \mathrm{cov}(Y,Z)$$

$$\mathrm{var}(R_2 - R_1) = \mathrm{var}(R_1) + \mathrm{var}(R_2) - 2\,\mathrm{cov}(R_2, R_1)$$

$$= \sigma_1^2 + \sigma_2^2 - 2\sigma_{12}$$

$$\mathrm{cov}(R_2 - R_1, R_2) = \mathrm{cov}(R_2, R_2) - \mathrm{cov}(R_1, R_2)$$

$$= \mathrm{var}(R_2) - \mathrm{cov}(R_1, R_2)$$

$$= \sigma_2^2 - \sigma_{12}$$

$$w = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} = \frac{\mathrm{cov}(\overbrace{R_2 - R_1}^{X}, \overbrace{R_2}^{Y})}{\mathrm{var}(\underbrace{R_2 - R_1}_{X})}$$

Same thing

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$Y = \beta_0 + \beta_1 X + \varepsilon$	$R_2 = \beta_0 + \beta_1 (R_2 - R_1) + \varepsilon$
$\hat{\beta}_1 = \dfrac{\mathrm{cov}(Y,X)}{\mathrm{var}(X)}$	$\hat{\beta}_1 = w$

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